EF-10

TEACHER GUIDE

Engineering Foundations · Programming Robots

Talking Chips: I2C, UART and SPI

A robot is a team of chips chatting. What language do they speak?

1 At a glance

LENGTH	TYPE	GROUPING	CONCEPTS
60 minutes	Software	Pairs (two pairs join up for the UART step)	serial · I2C addresses · UART · protocols

2 Overview

Students learn why a robot's chips share wires, and the three agreements (protocols) that make sharing work. They put two I2C devices on the same two wires, find their addresses with a scanner, show a sensor reading on a screen, then make two Arduinos talk over UART. The lesson ends with a rule of thumb for choosing I2C, UART or SPI that they will use in every later build.

> **The one idea that must land**
>
> Wires are just wires; the protocol is the agreement. I2C shares two wires by giving every device an address. UART joins exactly two boards, TX to RX, at an agreed speed. And every link needs a shared ground.

3 Learning outcomes

By the end, students can:

1. Explain what **serial** communication is, and why robots use it instead of parallel.
2. Put **two I2C devices** on the same two wires and find their addresses with a scanner.
3. Send data between **two boards over UART**, crossing TX to RX and matching the baud rate.
4. Choose **I2C, UART or SPI** for a job and give a reason.

4 Three depths

Pick one for the class. Everyone does the same lesson; only the build target changes.

Starter

Pass-the-message game: serial versus parallel

Run the Be the bus game and the pass-the-message comparison of serial and parallel, then use the scanner on a ready-wired I2C bus. Stop after build step 1. Younger groups can do the whole build in Circuit Lab.

Core

OLED and sensor on one I2C bus, scanned and used

Full lesson: wire the OLED and MPU6050 on one bus, scan it, show the tilt on the screen (steps 1 and 2), then the two-board PING if time allows.

Advanced

Two-board messaging protocol with a checksum

Do all three steps quickly, then design a tiny message format with a checksum: send `T,23,5` and have board B add up the numbers and reject a message that does not match.

5 Before the lesson

Kit, per group of 2

- Arduino Uno + USB cable (two per pair for the UART step, so pairs join up)
- OLED 0.96 in I2C display (address 0x3C)
- MPU6050 motion sensor module (address 0x68)
- Half-size breadboard and male-male jumper wires
- One push button and one LED with a 220 Ω resistor (for the UART step; or use the Uno's built-in LED on pin 13)

Simulators

- **Circuit Lab**: both I2C modules and two-board UART can be built and run on screen, so a pair without kit loses nothing.
- **Wokwi** (online): a second option with a ready OLED example.

Prep checklist

- [ ] Install the **Adafruit SSD1306** and **Adafruit GFX** libraries on every laptop (Library Manager). Step 2 will not compile without them.
- [ ] Load the scanner sketch on one board and check both modules answer (0x3C and 0x68). Swap out any dead module now.
- [ ] Check every USB cable carries data, not just charge: the board should appear as a port.
- [ ] Prepare sticky notes numbered 1 to 30, with two notes carrying the same number, for the Be the bus game.
- [ ] Decide which pairs join up for the UART step, and have the two sketches ready to share.

6 Timing plan

Min	Phase	What happens
5	Let's think	Why chips must share wires; students write three guesses (slide 2).
3	Aims	The four aims and the keywords (slide 3).
17	Teach	Serial vs parallel, I2C and addresses, UART, SPI, the comparison table and real examples (slides 4 to 10).
5	Be the bus	The address game, then round two with a clapped clock (slide 12).
7	Wire and scan	Wire both modules to A4/A5, run the scanner, record the addresses (slides 13, 14, 16).
8	Both devices, one sketch	Tilt from the MPU6050 shown on the OLED (slide 17).
7	Two boards talking	Pairs join; crossed wiring, PING, LED blinks (slides 15, 18).
3	Break it on purpose	Two or three faults from the test table (slide 19).
5	Check and exit ticket	Self-rating against the proofs; GPS, screen, SD card question (slides 21, 22).

Total: 60 minutes

7 Step by step

Let's think

5 min · slides 1-2

Say: An Arduino has 20 pins. Count the parts on a real robot: screen, motion sensor, clock, SD card, radio. Do we have enough pins?

Say: So the chips have to share wires. What rules would they need?

→ Students write three guesses: how few wires could carry a sentence; what happens if two chips talk at once; how a chip knows a message is for it.

→ Keep the guesses. You will come back to them in the exit discussion.

Look for: students who say 'more pins' or 'a bigger board'. That is a fair answer; ask what happens on a robot with 40 parts.

Aims

3 min · slides 3

Say: Today you will make two chips share two wires, and two whole Arduinos talk to each other.

→ Read the four aims. Point at the keywords note: SDA, SCL, TX, RX, baud rate will all be used today.

Teach

17 min · slides 4-10

Say: Parallel sends eight bits at once on eight wires. Serial sends them one after another on one wire. Nearly everything modern is serial: USB, HDMI, Ethernet.

Say: I2C uses two wires. SDA carries the data. SCL is the clock: a bit is only read on a tick, so nobody loses their place.

Say: Every I2C device has an address, like a house number. The Arduino shouts the number; only that device answers. Our OLED is 0x3C and the MPU6050 is 0x68.

Say: UART has no clock and no addresses. Two boards agree a speed in advance, the baud rate, and TX of one goes to RX of the other.

Say: SPI is the fast one, for SD cards and big displays, but it needs four wires plus one extra pin per device.

- → Slide 6: show that 0x3C is 60 in decimal and 0011 1100 in binary. One hex digit is four bits.
- → Slide 9: read the rule of thumb aloud. A few slow devices sharing wires: I2C. Two boards chatting: UART. Something fast: SPI.
- → Slide 10: ask for one more example from home for each protocol.

Look for: the idea that TX goes to TX. Draw it on the board: what one board says (TX) must arrive at the other board's ear (RX).

Be the bus

5 min · slides 12

Say: You are all devices on one bus. Your sticky note is your address. I am the master.

Say: Round two: you may only speak when I clap. My clap is SCL, the clock.

- → Call a number and a short message; only that student replies.
- → Call the duplicated number. Two students answer at once and nobody hears either. That is an address clash.
- → In round two, clap a slow steady beat; students say one word per clap.

Look for: students noticing that the clap keeps everyone in step. That is exactly why I2C has a clock wire and UART needs an agreed baud rate instead.

Wire and scan

7 min · slides 13, 14, 16

Say: Both modules go on the same four rails: 5V, GND, SDA to A4, SCL to A5. That is the whole point. Ground first, every time.

- → Pairs wire both modules from the Wiring slide (14).
- → Upload the scanner, open the Serial Monitor at 9600 baud, write down every address found.
- → Expect `Found 0x3C` and `Found 0x68`. Some MPU6050 boards report 0x69 if their AD0 pin is high.

Look for: an empty scan. In order, check: ground joined, SDA and SCL not swapped, module powered (its LED is on).

Both devices, one sketch

8 min · slides 17

Say: Two libraries, two addresses, one pair of wires. Neither device knows the other is there.

- → Upload the step 2 sketch. Tilt the breadboard: the number on the OLED should change.
- → Ask pairs to unplug one module and see what the other does.

Look for: a blank OLED with a working scan. The address in `oled.begin` must match what the scanner found.

Two boards talking

7 min · slides 15, 18

Say: Pin 11 of A goes to pin 10 of B, and pin 11 of B goes to pin 10 of A. Then join the grounds. This is not optional.

→ Two pairs join up: one board runs the sender, the other the receiver.

→ Press the button on board A; the LED on pin 13 of board B blinks.

→ We use SoftwareSerial on pins 10 and 11 so pins 0 and 1 stay free for USB uploads.

Look for: boards that 'do nothing'. Silence usually means wiring (TX to TX, or no shared ground); gibberish means the baud rates differ.

Break it on purpose

3 min · slides 19

Say: An engineer who can describe the symptom can find the fault. 'It does not work' is not a symptom.

→ Each pair tries two rows of the table and writes down what they saw.

Check and exit ticket

5 min · slides 21-22

Say: Rate yourself red, amber or green on each aim, and be ready to show the proof.

→ Exit ticket: a GPS, a screen and an SD card on one robot. Which protocol for each, and why?

→ Return to the three guesses from the start and ask who would now answer differently.

8 Common misconceptions

Students may think	How to help
TX connects to TX, like for like.	What one board transmits must reach the other's receiver. Act it out: a mouth speaks into an ear, not into another mouth.
Each I2C device needs its own pins.	That is what I2C avoids. Both modules share A4 and A5; the address picks which one answers.
0x3C is a strange code to memorise.	It is just the number 60 written in base 16. The scanner always tells you the real address, so nobody needs to memorise it.
If it gets 5 V it does not need ground.	Voltage is always measured against something. Without a shared ground the signals have no reference and mean nothing.
Faster is always better, so use SPI.	SPI costs a pin per device. For a few slow sensors, I2C's two wires are the better engineering choice.

9 Troubleshooting

Problem	Fix
Scanner prints nothing	Check GND, then swap SDA and SCL, then check the module's power LED. Re-seat the jumpers in the breadboard.
Scanner finds 0x69, not 0x68	The MPU6050's AD0 pin is pulled high. Use 0x69 in the sketch, or tie AD0 to GND.
OLED stays blank	Use the address the scanner found (some screens are 0x3D). Check the SSD1306 library is installed and the size is 128×64.

Step 2 will not compile	Install Adafruit SSD1306 and Adafruit GFX from the Library Manager.
UART receiver shows gibberish	Both sketches must use the same baud rate (9600).
UART receiver shows nothing	Pin 11 must go to pin 10 on the other board in both directions, and the two GNDs must be joined.
Upload fails while boards are linked	Unplug the wire on the receiving pin during upload, or check you are not using pins 0 and 1.

10 Assessment rubric

	Emerging	Secure	Extending
Explaining serial and protocols	Says serial uses fewer wires, with prompting.	Explains serial vs parallel and why a protocol is needed, unprompted.	Explains why serial won (cleaner, faster single line) with an example.
Building the I2C bus	Wires one module with help; scanner finds it.	Both modules on one bus; scanner finds both; tilt shows on the OLED.	Adds a third device or fixes an address clash without help.
UART and choosing a protocol	Gets a message across with help.	Crosses TX and RX, shares ground, explains gibberish vs silence.	Chooses the right protocol for new parts and justifies it; designs a checksum.

11 Answers

Slide 2: How few wires could carry a whole sentence?

> One data wire plus ground (UART, one direction). I2C uses two plus ground for many devices.

Slide 2: If both chips talk at once, what happens?

> Neither message gets through (a clash). Protocols prevent this with addresses (I2C) or by joining only two devices (UART).

Slide 2: How does a chip know a message is for it?

> In I2C, each message starts with the device's address; only the matching device answers.

Slide 6: What is 0x3C in decimal and binary?

> 60, and 0011 1100.

Slide 19: Set board B to 4800 baud. What happens?

> Gibberish appears instead of PING. Wrong speed gives nonsense; wrong wiring gives silence.

Exit ticket: GPS, screen and SD card on one robot.

> GPS: UART (it streams text on its own TX line). Screen: I2C (slow, shares two wires with other sensors). SD card: SPI (lots of data, needs speed).

12 Safety

- ! Low voltage only: everything runs from USB.
- ! Unplug USB before changing wiring, so a misplaced jumper cannot short 5V to GND.
- ! If a module gets hot to the touch, unplug at once: it is wired backwards (VCC and GND swapped).
- ! When two boards are linked, always join their grounds before connecting the signal wires.